

Credit Card Transactions

Data Analysis Case Study using MS SQL Server

1. Write a query to print top 5 cities with highest spends and their percentage contribution of total credit card spends

Results		Messages	
	city	TotalSpend	PercContribution
1	Greater Mumbai	576751476.00	14.15
2	Bengaluru	572326739.00	14.05
3	Ahmedabad	567794310.00	13.93
4	Delhi	556929212.00	13.67
5	Kolkata	115466943.00	2.83

2. Write a query to print highest spend month and amount spent in that month for each card type

Results		Messages		
	card_type	txn_month	txn_year	TotalSpend
1	Gold	1	2015	55455064
2	Platinum	8	2014	57936507
3	Signature	12	2013	58799522
4	Silver	3	2015	59723549

3. Write a query to print the transaction details (all columns from the table) for each card type when it reaches a cumulative of 1000000 total spends

Results		Messages					
	transaction_id	city	transaction_date	card_type	exp_type	gender	amount
1	1521	Delhi	2013-10-04	Gold	Food	M	281924
2	190	Ahmedabad	2013-10-05	Platinum	Bills	F	612572
3	72	Delhi	2013-10-04	Signature	Bills	F	550782
4	7564	Bengaluru	2013-10-04	Silver	Food	F	205179

4. Write a query to find city which had lowest percentage spend for Id card type

Results		Messages		
	city	gold_spend	city_total_spend	gold_spend_perc
1	Dhamtari	1416	425241	0.33

5. Write a query to print 3 columns: city, highest_expense_type, lowest_expense_type

Results		Messages	
	city	highest_expense_type	lowest_expense_type
13	Akola	Bills	Fuel
14	Akot	Fuel	Entertainment
15	Alappuzha	Food	Entertainment
16	Aligarh	Bills	Entertainment
17	Alipurduar	Food	Entertainment
18	Alirajpur	Entertainment	Entertainment
19	Allahabad	Grocery	Bills
20	Alwar	Food	Entertainment

6. Write a query to find percentage contribution of spends by females for each expense type

Results Messages		
	exp_type	contribution_by_females
1	Entertainment	49.37
2	Food	54.91
3	Bills	63.95
4	Fuel	49.71
5	Travel	51.13
6	Grocery	50.91

7. Which card and expense type combination saw highest month over month growth in Jan-2014

Results Messages							
	card_type	exp_type	trans_month	trans_year	spend	prev_month_spend	MoM_growth_perc
1	Gold	Travel	1	2014	2092554	1113534	87.92

8. During weekends which city has highest total spend to total no of transactions ratio

Results Messages				
	city	total_weekend_spend	total_weekend_txn	spend_per_txn_ratio
1	Sonepur	299905	1	299905.00

9. Which city took least number of days to reach its 500th transaction after the first transaction in that city

Results Messages		
	city	day_to_500th_transaction
1	Bengaluru	81